Charles's law

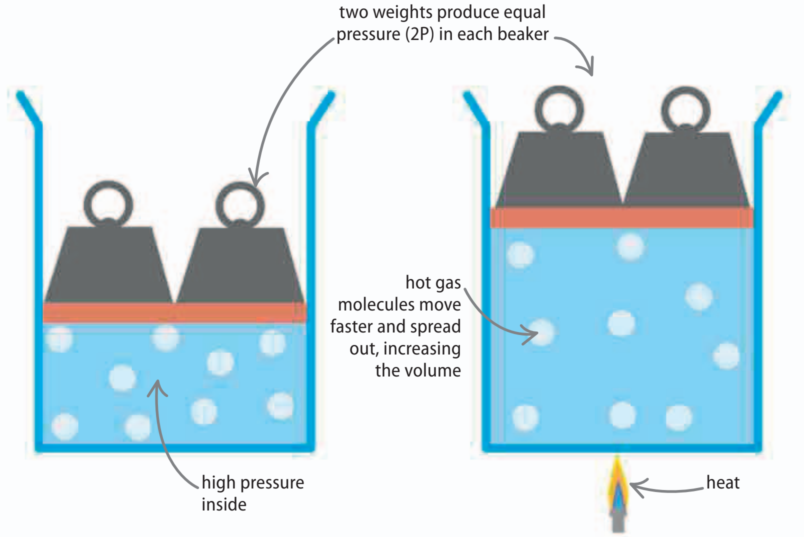
This gas law, which is attributed to the French scientist Jacques Charles, states that the temperature of a gas is proportional to its volume. So if the gas is held in a container with an adjustable volume—a gas syringe, for example—increasing the temperature of the gas results in an increase in its volume.

V stands for "volume"

T stands for "temperature"

$$V \propto T$$

△ **Equation for Charles's law**
This equation shows the relationship between a gas's volume and its temperature. Increasing the temperature increases the volume.



△ **Temperature**
Temperature is a measure of heat energy: the motion of a gas's molecules. Increasing the temperature of the gas increases the rate at which its molecules move.

△ **More motion**
Faster molecules hit each other and the container walls more often. If one wall is moveable, these impacts will push it outward, increasing the volume of the container.

Gay-Lussac's law

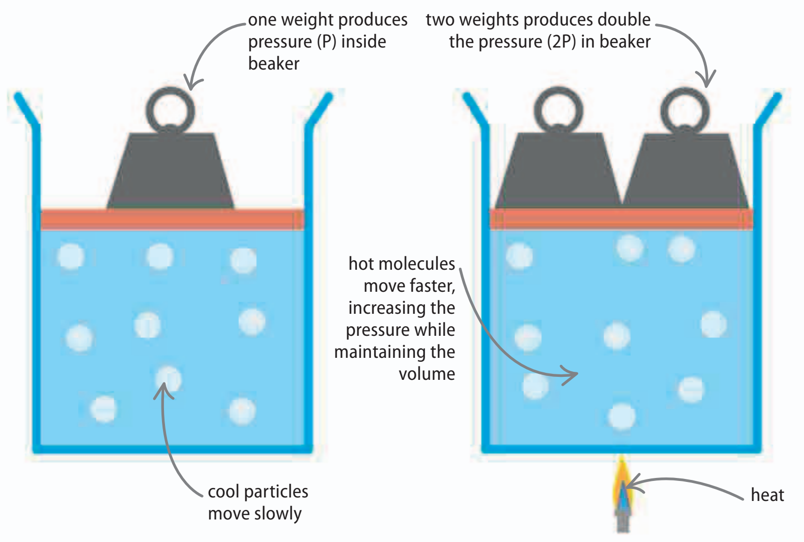
Named after French scientist Joseph Louis Gay-Lussac in 1808, this was the last of the three main gas laws to be formulated. It states that for a fixed volume of gas, the pressure is proportional to its temperature. In other words, when the temperature of a gas is increased, it also exerts a higher pressure. Similarly, squeezing a gas into a smaller volume increases its pressure (as per Boyle's law) and also raises the gas's temperature.

P stands for "pressure"

T stands for "temperature"

$$P \propto T$$

△ **Equation for Gay-Lussac's law**
This equation shows the relationship between the pressure of a gas and its temperature. Increasing the temperature increases the pressure.



△ **Fewer collisions**
The molecules in the cool gas move slowly and they hit the sides of the container infrequently. These few, weak collisions combine to create a low gas pressure, overall.

△ **More collisions**
As the gas is heated, the molecules move around faster and hit the sides of the container more often and with greater force. Thus the pressure goes up.